Phase 2 of the Brown–Zudilin $\zeta(5)$ denominator law: reduction to a single period connection coefficient

20 July 2026

Abstract

The corrected denominator law $12\,d_n^5 P_n \in \mathbb{Z}$ for Brown–Zudilin's totally symmetric cellular approximations to $\zeta(5)$ splits into two regimes. The large primes $n < p \le 2n$ (Phase 1) are a theorem: the Frobenius certificate, now also kernel-verified in Lean 4. This note reports the state of the small primes $p \le n$ (Phase 2). We prove that the entire $p \ge 5$ law is equivalent to a single one-step descent (D); that (D) reduces, via the Apéry-like recurrence shared by the coefficient triple $(Q_n, P_n, \widehat{P}_n)$, to finitely many *singular gates* per base-p digit block; and that the generic *midpoint gate* at $a = 1$ is a theorem, through an explicit certificate chain whose non-vanishing input is supplied by the exact Casoratian. We then prove a precise *impossibility* result: the remaining *cubic gate* cannot be closed by the recurrence and Casoratian alone, because a determinant-one change of basis preserves both while altering which solution is regular. The whole difficulty is thereby localized to one transcendental input — the P-column of the one-digit Frobenius connection matrix, a vector-valued Dwork congruence for these harmonic companion sequences — which we pose as the open problem. All computational claims are exact; the adversarial evidence base for (D) stands at $n \le 360$, $p \le 73$, 5,989 descents, zero violations.

1 Setup and the corrected law

For $n \ge 1$ let

$$R_n(k) = (n!)^4 \Big(k + \tfrac{n}{2}\Big) \frac{\prod_{j=1}^{n}(k-j)\,\prod_{j=1}^{n}(k+n+j)}{\prod_{j=0}^{n}(k+j)^6},$$

Zudilin's totally symmetric rational function [1]. Its partial fractions define the functionals u, w, v and their companions $\widetilde{u}, \widetilde{w}, \widetilde{v}$ (obtained from $-k(k+n)R_n$), and the coefficient ladders

$$q_n = u\widetilde{w} - \widetilde{u}w, \qquad p_n = \widetilde{w}v - w\widetilde{v}, \qquad \widetilde{p}_n = u\widetilde{v} - \widetilde{u}v.$$

The Brown–Zudilin normalized coefficients are $Q_n = (-1)^{n+1} q_n / \binom{2n}{n}$, $P_n = (-1)^{n+1} p_n / \binom{2n}{n}$, $\widehat{P}_n = (-1)^{n+1} \widetilde{p}_n / \binom{2n}{n}$; Q_n is a positive integer (the double-binomial period), and the target is

$$12\, d_n^5\, P_n \in \mathbb{Z}, \qquad d_n = \operatorname{lcm}(1, \dots, n), \tag{1}$$

with the constant $12 = 2^2 \cdot 3$ conjecturally sharp. Brown and Zudilin state (1) without the 12 as an experimental observation; that uncorrected form fails already at $n = 2$ (their displayed $P_2 = 1190161/384$ gives $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$), and (1) is the sharp repair [2, 4].

Because the crude tier $d_{2n}^5 P_n \in \mathbb{Z}$ is elementary, a failure of (1) at a prime $p \ge 5$ can occur only when $p^k \in (n, 2n]$ for some $k \ge 1$. The case $k = 1$, i.e. $n < p \le 2n$, is Phase 1: $p \mid p_n$, proved by the Frobenius certificate $(k^p - k)^2$ and the residue theorem [2], and kernel-verified in Lean 4 [3]. This note concerns $p \le n$.

2 The master reduction to a one-step descent

Since $v_p(P_n) = v_p(p_n) - v_p\binom{2n}{n}$ and Kummer's theorem controls $v_p\binom{2n}{n}$ by base-p carries, the carries cancel identically in the P-normalization, leaving a single clean statement.

Definition 1 (the descent (D))**.**

$$\text{(D)}: \qquad v_p(P_n) \geq v_p\big(P_{\lfloor n/p \rfloor}\big) - 5 \qquad (p \geq 5,\ p \leq n).$$

Theorem 2 (master reduction)**.** (D) *implies* $v_p(P_n) \geq -5\lfloor \log_p n \rfloor$ *for every* n *and every prime* $p \geq 5$ *— equivalently, the* p*-part of the corrected law* (1) *at all primes* $p \geq 5$*.*

Proof. Iterate (D) down the digit chain $n \mapsto \lfloor n/p \rfloor \mapsto \cdots \mapsto n_L$ with $n_L < p$, $L = \lfloor \log_p n \rfloor$, giving $v_p(P_n) \geq v_p(P_{n_L}) - 5L$. At the terminal index either $2n_L < p$, whence $P_{n_L} \in \mathbb{Z}_p$ by Zudilin's denominator theorem (neither d_{n_L} nor $\binom{2n_L}{n_L}$ meets p), or $n_L < p \leq 2n_L$, whence Phase 1 supplies the one missing power of p. In both cases $v_p(P_{n_L}) \geq 0$. $\square$

Theorem 2 is formalized in Lean 4 as `descent_law`, with (D) an explicit hypothesis; the $\{2,3\}$-part of the constant 12 is a separate finite obligation [3]. Thus the whole law reduces to (D).

3 The gate structure

The triple $(Q_n, P_n, \widehat{P}_n)$ satisfies one Apéry-like third-order recurrence [1]. In the $\binom{2n}{n}$-normalization it reads $\sum_{i=0}^{3} c_i(m) Y_{m+i} = 0$ with, crucially,

$$c_3(m) = 2(m+3)^5(2m+5)\, a_0(m), \qquad a_0(m) = 41218m^3 + 198849m^2 + 320790m + 173057. \tag{2}$$

The fifth power $(m+3)^5$ is intrinsic: it is the exact per-digit loss of (D), not a fitted constant. Reducing the recurrence over a discrete valuation ring of residue characteristic $p \geq 5$ and writing $n = ap + r$, $0 \leq r < p$, one has $c_i(ap + r) \equiv c_i(r) \pmod p$, so on a digit block the recurrence has unit leading coefficient and propagates p-integrality except at the residues where

$$(2r + 5)\, a_0(r) \equiv 0 \pmod p.$$

The boundary factor $(r + 3)^5$ contributes only the sanctioned five-order loss at $r = p - 3$.

Proposition 3 (finite-gate lemma)**.** *Each base-*p *digit block reduces to ordinary integral propagation plus at most four* singular gates*: one* midpoint *gate at the root of* $2r + 5$*, and at most three* cubic *gates at the roots of* a_0*. The gates are represented by explicit small-integer certificate rows: an exact* $\mathbb{Z}[m]$ *identity*

$$128\, c_i(m) = 7R_i + (2m + 5)H_i(m), \quad (R_0, R_1, R_2) = (11907, -334374, -19292),$$

for the midpoint gate, and $D\, c_i(m) = A_i(m) + a_0(m)K_i(m)$ *with* $D = 2^5 \cdot 37^5 \cdot 557^2$ *for the cubic gate. The exceptional primes are exactly* $\{7, 29, 107, 557, 673\}$ *(from* $\operatorname{disc} a_0$*) and* $\{43, 701\}$ *(from* $\operatorname{Res}_m(a_0, 2m + 5)$*).*

This is the per-(n, p)-witness-plus-uniform-identity shape needed both for a proof and for eventual formalization.

4 The $a = 1$ midpoint gate is a theorem

Fix an odd prime p outside the exceptional set, put $r = (p-5)/2$ and $N = p + r = (3p-5)/2$, so $2N + 5 = 3p$. Write $M_p = 11907\, P_N - 334374\, P_{N+1} - 19292\, P_{N+2}$ and $e_p = \min_{0 \le s \le 2} v_p(P_{N+s})$.

Theorem 4 ($a = 1$ midpoint gate)**.** *$v_p(M_p) \ge e_p + 1$; equivalently the midpoint gate at $a = 1$ holds in saturated form, $v_p(M_p) \ge -4$ with $e_p = -5$.*

The proof is an exact determinant assembly (DET-MID) in which neither regular residual is discarded before the determinant is formed, reduced through the head/middle/tail chambers of the pole index. It combines five established facts:

(i) the Q-coordinate vanishes mod p, from the machine-verified Lucas–Frobenius congruence $Q_{ap+r} \equiv Q_a Q_r$ and the midpoint row;

(ii) (RM) the middle regular determinant lies in $p^4\mathbb{Z}_p$, by a factorial block count;

(iii) (RT) the reflected head+tail regular determinant lies in $p\mathbb{Z}_p$ — an eight-order cancellation whose inputs are individually non-integral, proved by an exact Φ/Bell substitution reducing it to three reflection-jet identities (T1)–(T3) in the harmonic alphabet (certificate "J12"), with the boundary pair and central collision handled as explicit sub-cases;

(iv) (H) the five-digit head-window normal form, whose local coefficients are $(0, 0, 0, 0, -(a_{1,b} + a_{1,r-b})/84)$; the residual sum $\sum_b C_4(b) \equiv 0 \pmod p$ is *exactly* the first infinity relation $E_1 = 0$, so (H) needs no new input;

(v) (Q3) the non-vanishing $\neg(Q_r \equiv Q_{r+1} \equiv Q_{r+2} \equiv 0)$.

Lemma 5 ((Q3), via the Casoratian)**.** *For every prime $p \ge 11$, not all of Q_r, Q_{r+1}, Q_{r+2} vanish mod p.*

Proof. The fundamental matrix of $(Q, P, \widehat{P})$ at rows $(n, n+1, n+2)$ has Casoratian

$$\det \mathcal{Y}_n = \frac{(-1)^{n-1} a_0(n)}{16\,(n+1)^4 (n+2)^5 (2n+1)(2n+3)\binom{2n}{n}}, \tag{3}$$

uniform from $\det \mathcal{Y}_{n+1} / \det \mathcal{Y}_n = -c_0/c_3$ and the base value $\det \mathcal{Y}_1 = 13591/34560$. At $n = r = (p-5)/2$ every denominator factor is a p-unit and $2r < p$ makes $\binom{2r}{r}$ a unit, so by the integrality lemma the P and $\widehat{P}$ columns are p-integral and $\det \mathcal{Y}_r$ is a p-unit *unless* $p \mid a_0(r)$. An all-zero Q-column would force $\det \mathcal{Y}_r \equiv 0$, a contradiction. Finally $8a_0((p-5)/2) = 41218p^3 - 220572p^2 + 397530p - 241144$ with $241144 = 8 \cdot 43 \cdot 701$, so $p \mid a_0(r)$ only for $p \in \{43, 701\}$, handled by direct double-binomial certificates ($[Q_{19}, Q_{20}, Q_{21}] \equiv [33, 0, 26] \bmod 43$; $[Q_{348}, Q_{349}, Q_{350}] \equiv [472, 350, 182] \bmod 701$). $\square$

The non-vanishing input is thus the recurrence's Casoratian, not a closed form: the corresponding half-period modular coefficient *can* vanish mod p (e.g. $Q_{(p-1)/2} \equiv 0 \bmod 19$), so only the determinantal argument succeeds.

5 The obstruction at the cubic gate

The cubic gate — regularity of P at a root of a_0 — does not yield to the same method, and the failure is structural.

Theorem 6 (impossibility for recurrence + Casoratian)**.** *No argument using only the scalar recurrence and its Casoratian can prove the cubic gate for P. Concretely, the determinant-one substitution*

$$(Q_n, P_n, \widehat{P}_n) \longmapsto (Q_n,\ P_n + \widehat{P}_n,\ \widehat{P}_n)$$

preserves both the recurrence and the Casoratian (3)*, yet exchanges the regularity of the second column at a root of a_0. Hence these data do not distinguish P from $P + \widehat{P}$.*

At a simple root of a_0 the reduced fundamental matrix has rank two; its adjugate is rank one and satisfies the explicit first-order recurrence $c_3(n)\,\mathrm{adj}\,\mathcal{Y}_{n+1} = \mathrm{adj}\,\mathcal{Y}_n\,\Lambda$ with Λ the companion block. The adjugate is regular exactly where the determinant vanishes — the classical Cramer/connection mechanism — but its first jet identifies only the regular two-plane, and the saturated null direction is genuinely mixed (e.g. $(30, 9, 30)$ at $(p, r) = (43, 19)$), not the pure $\widehat{P}$ axis. The exact oracle over every a_0-root prime below 50 confirms P is *ordinarily* regular there (no $\widehat{P}$-subtraction is needed), but that is a statement the toolkit cannot prove.

6 The remaining problem

Everything now rests on one input. Writing ℓ_r for the cubic certificate row:

Problem 7 (the vector Dwork connection coefficient)**.** Prove

$$\ell_r\big(p^5 P_N,\ p^5 P_{N+1},\ p^5 P_{N+2}\big) \equiv 0 \pmod{p},$$

equivalently determine the P-column of the one-digit Frobenius connection matrix relating the period vector at $n = ap + r$ to that at $\lfloor n/p \rfloor = a$. This is a vector-valued Dwork congruence for the harmonic companion sequences $(Q, P, \widehat{P})$ and must be read off the hypergeometric/period determinant, not the recurrence.

Several independent lines converge on Problem 7: a linear $\mathbb{Z}/p^2$-span experiment shows the actual gain is bilinear, invisible to any linear certificate; the salvage of the primal Barnes integral collapses the symmetric family to an order-two kernel $\mathcal{R}_n(s,t) = n!\,(s{+}1)_n(t{+}1)_n(s{+}t{+}n{+}2)_n/[(s{+}n{+}1)^2_{n+1}(t{+}n{+}1)^2_n$ times a universal sine kernel, which is the natural home for such a connection coefficient; and the impossibility Theorem 6 shows no recurrence-only route can substitute for it. A general matrix Dwork theory provides the architecture, but no off-the-shelf theorem covers these particular sequences.

7 Evidence and formalization status

Evidence. All coefficient data are exact (Fraction/$\mathbb{Z}$), extended to $n = 360$ by the recurrence and cross-validated against direct partial-fraction computation at 151 points. An adversarial sweep found no violation of (1) or (D) over $n \le 360$, $p \le 73$: 5,989 descents, zero failures, including the previously untested three-digit descents at $p = 5, 7$. The law rides the boundary — slack 0 at $p = 2$ for every n, so the constant 12 is exactly sharp throughout. This is evidence, not proof.

Formalization (Lean 4, kernel-verified, standard axioms). Proved and machine-checked: the erratum with its sharp repair; the Phase 1 certificate theorem $p \mid p_n$ for $n < p \le 2n$; the corrected law $12 d_n^5 P_n \in \mathbb{Z}$ for $n \le 8$ with non-vanishing; the Lucas–Frobenius row $Q_{ap+r} \equiv Q_a Q_r$; and the master reduction Theorem 2 as the conditional theorem (D) $\Rightarrow$ full law. The $a = 1$ midpoint gate (Theorem 4) and the impossibility (Theorem 6) are proved on paper with every step exact-verified, and are natural next formalization targets once the cubic gate is understood.

Summary. Phase 2 is reduced — unconditionally, through a chain of exact and in part machine-checked steps — to Problem 7, a single period connection coefficient. The recurrence-based toolkit is provably exhausted at that point; closing it requires the period determinant.

Acknowledgments

The Phase 2 reduction reported here — the master descent (D), the finite-gate structure, the Φ/Bell certificate chain closing the $a = 1$ midpoint gate (J12, (RT), (H), (H-NF)$= E_1$, and (Q3) via the Casoratian), and the impossibility Theorem 6 localizing the whole difficulty to Problem 7 — is due in its mathematical substance to Sol (a GPT-5.6 instance), across a sustained multi-session collaboration. Each step was exact-verified before being recorded, and the negative results (the falsified clean-diagonal Frobenius form, the exhausted recurrence toolkit) were established with the same rigor as the positive ones.

References

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